1 Supplement: linear algebra in GNSS

This supplement connects four topics of an advanced linear algebra course to computations that recur in GNSS positioning: vector and matrix norms, Hermitian matrices (real symmetric, in estimation), the singular value decomposition (SVD), and the Moore–Penrose pseudoinverse. Thirteen numerical examples carry the argument. All of them use one simulated epoch (Section 1.1), and every number is printed by `scripts/linalg_supplement_examples.py` (Section 1.7). Where the engine's own version of a result is derived elsewhere in this reference, the text links to it.

1.1 The model used throughout

Linearised GNSS observation equations have the form

$$\boldsymbol{y} = \boldsymbol{A}\,\boldsymbol{x} + \boldsymbol{\varepsilon}, \qquad \mathsf{E}\{\boldsymbol{\varepsilon}\} = \boldsymbol{0}, \qquad \mathsf{D}\{\boldsymbol{\varepsilon}\} = \boldsymbol{R}, \tag{1}$$

with m observations, n parameters and observation covariance $\boldsymbol{R}$. For code positioning the design matrix is the geometry matrix $\boldsymbol{G}$, with one row per satellite,

$$\boldsymbol{g}_i^{\mathrm{T}} = \begin{bmatrix} -\boldsymbol{e}_i^{\mathrm{T}} & 1 \end{bmatrix}, \tag{2}$$

where $\boldsymbol{e}_i$ is the unit line of sight to satellite i, expressed here in local east, north and up components, and the last column belongs to the receiver clock $c\,\mathrm{d}t_r$, kept in metres (the linearised geometric range of the Foundations document). One epoch of code and phase on a single frequency adds one float ambiguity N_i, in cycles, per satellite:

$$P_i = \boldsymbol{g}_i^{\mathrm{T}}\boldsymbol{x} + \varepsilon_i^{P}, \qquad \Phi_i = \boldsymbol{g}_i^{\mathrm{T}}\boldsymbol{x} + \lambda_{\mathrm{L1}} N_i + \varepsilon_i^{\Phi}, \qquad i = 1, \ldots, m, \tag{3}$$

with P_i and Φ_i the prefit residuals in metres and $\boldsymbol{x}$ the corrections to east, north, up and the clock. Weighted least squares with $\boldsymbol{W} = \boldsymbol{R}^{-1}$ gives the best linear unbiased estimator and its covariance,

$$\hat{\boldsymbol{x}} = \boldsymbol{N}^{-1}\boldsymbol{A}^{\mathrm{T}}\boldsymbol{W}\boldsymbol{y}, \qquad \boldsymbol{P}_{\hat{x}} = \boldsymbol{N}^{-1}, \qquad \boldsymbol{N} = \boldsymbol{A}^{\mathrm{T}}\boldsymbol{W}\boldsymbol{A}, \tag{4}$$

where $\boldsymbol{N}$ is the normal matrix. Figure 1 shows the simulated epoch, and the table below lists its settings.

Setting	Value
Signal	GPS L1, $\lambda_{\mathrm{L1}} = 0.1903$ m
Azimuth / elevation	S1 15°/62°, S2 70°/25°, S3 120°/40°, S4 160°/15°, S5 210°/55°, S6 250°/30°, S7 300°/18°, S8 340°/80°
Code σ_P	0.3 m (geodetic receiver); 3 m (low-cost receiver, Example 11)
Phase σ_Φ	3 mm (geodetic receiver); 2 mm (low-cost receiver)

1.2 Norms

Four norms and their unit balls.

$$\|\boldsymbol{v}\|_1 = \sum_i |v_i|, \qquad \|\boldsymbol{v}\|_2 = \Big(\sum_i v_i^2\Big)^{1/2}, \qquad \|\boldsymbol{v}\|_\infty = \max_i |v_i|, \qquad \|\boldsymbol{v}\|_{\boldsymbol{W}} = \big(\boldsymbol{v}^{\mathrm{T}}\boldsymbol{W}\,\boldsymbol{v}\big)^{1/2}, \tag{5}$$

where $\boldsymbol{W}$ is symmetric positive definite. The first three treat every component alike. The fourth builds its metric from a matrix, and in estimation that matrix is an inverse covariance (Figure 2).

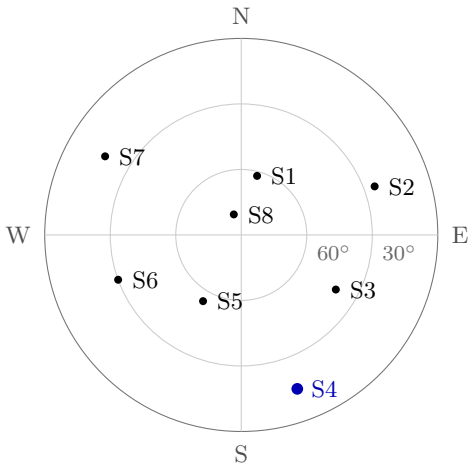


Figure 1: Sky plot of the eight satellites used in every example: azimuth clockwise from north, zenith at the centre, horizon at the rim. S4, at 15° elevation, carries the multipath error of Example 1. No satellite is below the horizon, which is the geometric reason for Example 8.

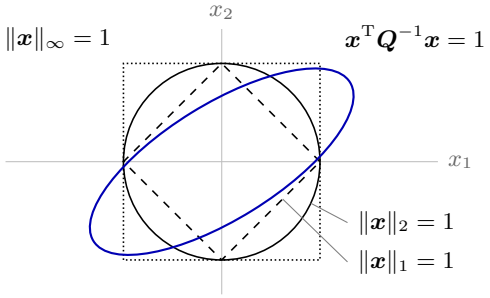


Figure 2: Unit balls of the four norms of Equation (5) in the plane: the L1 diamond (dashed), the L2 circle (solid), the L∞ square (dotted) and, in blue, the weighted norm with $\boldsymbol{W} = \boldsymbol{Q}^{-1}$ for $\boldsymbol{Q} = \left[\begin{smallmatrix} 1.8 & 0.9 \\ 0.9 & 0.9 \end{smallmatrix}\right]$. Its unit ball is an ellipse with axes along the eigenvectors of $\boldsymbol{Q}$ and semi-axes $\sqrt{\lambda_i(\boldsymbol{Q})}$; whitening by the Cholesky factor of $\boldsymbol{Q}$ maps it onto the circle.

Least squares picks its norm from the covariance. Weighted least squares minimises a weighted L2 norm of the residuals,

$$\hat{\boldsymbol{x}} = \arg\min_{\boldsymbol{x}} \|\boldsymbol{y} - \boldsymbol{A}\boldsymbol{x}\|^2_{\boldsymbol{R}^{-1}}, \tag{6}$$

so choosing the weight matrix is choosing the norm. With the Cholesky factor $\boldsymbol{R} = \boldsymbol{L}\boldsymbol{L}^{\mathrm{T}}$, the whitened quantities $\bar{\boldsymbol{y}} = \boldsymbol{L}^{-1}\boldsymbol{y}$ and $\bar{\boldsymbol{A}} = \boldsymbol{L}^{-1}\boldsymbol{A}$ have unit covariance, and Equation (6) becomes ordinary least squares, $\min\|\bar{\boldsymbol{y}} - \bar{\boldsymbol{A}}\boldsymbol{x}\|_2^2$. The same squared norm returns throughout GNSS quality control:

- the overall model test behind RAIM, $T = \hat{\boldsymbol{\varepsilon}}^{\mathrm{T}}\boldsymbol{R}^{-1}\hat{\boldsymbol{\varepsilon}} \sim \chi^2(m-n)$ when the model is correct (the variance-factor section of the Stochastic Model and QC document);
- the Kalman innovation test, $\boldsymbol{\nu}^{\mathrm{T}}\boldsymbol{S}^{-1}\boldsymbol{\nu} \sim \chi^2(m_k)$ (the gating section of the Kalman filter document);
- integer least squares, $\check{\boldsymbol{N}} = \arg\min_{\boldsymbol{a}\in\mathbb{Z}^n} \|\hat{\boldsymbol{N}} - \boldsymbol{a}\|^2_{\boldsymbol{Q}^{-1}_{\hat{N}}}$ (the integer least-squares problem of the Ambiguity Resolution document), whose ratio test compares the two smallest values of this norm.

The residuals are a projection of the data,

$$\hat{\boldsymbol{\varepsilon}} = (\boldsymbol{I} - \boldsymbol{\Pi}_A)\,\boldsymbol{y}, \qquad \boldsymbol{\Pi}_A = \boldsymbol{A}\boldsymbol{N}^{-1}\boldsymbol{A}^{\mathrm{T}}\boldsymbol{W}. \tag{7}$$

For uncorrelated observations the diagonal entries r_i of $\boldsymbol{I} - \boldsymbol{\Pi}_A$ are the redundancy numbers of the Stochastic Model and QC document. They sum to $m - n$, and r_i is the share of an error in observation i that stays visible in its own residual.

L1: bounding the influence of one observation. The L2 norm squares residuals, so a single large error dominates the fit. Least absolute deviations, $\min\|\boldsymbol{y} - \boldsymbol{A}\boldsymbol{x}\|_1$, grow only linearly with the residual. Robust GNSS estimators sit between the two and run as iteratively reweighted least squares. The Huber function is quadratic near zero and linear in the tails,

$$\rho(v) = \begin{cases} \tfrac{1}{2}v^2, & |v| \le k, \\ k|v| - \tfrac{1}{2}k^2, & |v| > k, \end{cases} \qquad w(v) = \min\Big(1, \frac{k}{|v|}\Big), \tag{8}$$

with v the standardised residual and w the weight of the next iteration. The IGG-III and Danish schemes go further and drive the weight of large residuals toward zero.

Example 1 (a multipath error on a low satellite). S4 (15° elevation) delivers a non-line-of-sight signal with a +20 m range error; the other code errors are Gaussian with $\sigma = 0.3$ m. Over 4000 trials:

Estimator	RMS 3-D position error
Least squares, no fault	0.52 m
Least squares, with the fault	12.89 m
Huber, $k = 1.5$	1.15 m
L1 (least absolute deviations)	0.99 m
Least squares, S4 excluded	0.61 m

Least squares hides the fault. The redundancy number of S4 is $r_4 = 0.354$, so only 7.1 m of the 20 m error stays in its own residual; the rest moves the solution by $(-1.9,\ 7.7,\ 10.2)$ m in east, north and up and by 8.0 m in the clock. The low satellites have the smallest redundancy numbers (S7 at 18°:

0.258; S2 and S4: 0.354; S3 at 40°: 0.673). This fault is large enough for Baarda's w-test [1] to flag: the expected statistic of the diagonal w-test is $20\sqrt{r_4}/0.3 = 39.7$ against a critical value of 3.29. The minimal detectable bias of the MDB formula for S4 is, however, 2.1 m: a smaller multipath error passes the test and still biases the position. The robust estimators cut the error by a factor of 11 to 13 with no explicit detection step, and excluding S4 after detection returns 0.61 m.

L∞: rounding, integrity and convergence. Rounding each float ambiguity to its nearest integer treats the coordinates separately. It returns the integer vector nearest in the unweighted L2 norm, and its pull-in region, the set of float values that round to $\boldsymbol{a}$, is a unit cube, which is a ball of the L∞ norm:

$$\mathcal{R}_{\boldsymbol{a}} = \left\{\boldsymbol{x} : \|\boldsymbol{x} - \boldsymbol{a}\|_\infty \le \tfrac{1}{2}\right\}. \tag{9}$$

Integer least squares measures distance in the $\boldsymbol{Q}_{\hat{N}}^{-1}$ norm instead. Its pull-in regions are Voronoi cells, hexagons in two dimensions:

$$\mathcal{S}_{\boldsymbol{a}} = \left\{\boldsymbol{x} : \|\boldsymbol{x} - \boldsymbol{a}\|_{\boldsymbol{Q}_{\hat{N}}^{-1}} \le \|\boldsymbol{x} - \boldsymbol{u}\|_{\boldsymbol{Q}_{\hat{N}}^{-1}} \text{ for all } \boldsymbol{u} \in \mathbb{Z}^n\right\}. \tag{10}$$

The L∞ norm has two other everyday roles. Integrity monitoring bounds the largest position error over the fault modes it considers, and iterative solvers stop when $\|\Delta\boldsymbol{x}\|_\infty$ falls below a tolerance.

Example 2 (rounding versus integer least squares). Two float ambiguities with standard deviations of 0.30 and 0.26 cycles and a correlation of 0.96. Success rates from 10^6 Monte Carlo samples:

Covariance	Correlation	κ	Rounding	Integer LS
$\boldsymbol{Q}_{\hat{N}}$	0.96	50.1	0.900	0.957
$\boldsymbol{Q}_{\hat{z}} = \boldsymbol{Z}\boldsymbol{Q}_{\hat{N}}\boldsymbol{Z}^{\mathrm{T}}$	0.03	7.8	0.957	0.957

The transformation $\boldsymbol{Z} = \left[\begin{smallmatrix} 1 & -1 \\ 1 & -2 \end{smallmatrix}\right]$ has integer entries and determinant -1, so it maps integers to integers and preserves $\det \boldsymbol{Q}_{\hat{N}}$ (4.77×10^{-4} cycles^4 before and after). Integer least squares does not change under it; rounding improves until it matches (Figure 3). Rounding also cannot use the correlation: at correlations of 0.98 and 0.99 it stays at 0.903 and 0.904, while integer least squares rises to 0.973 and 0.989. The decorrelation of the LAMBDA section of the Ambiguity Resolution document makes the search space nearly spherical so that the integer search is fast [10]; it does not change the answer.

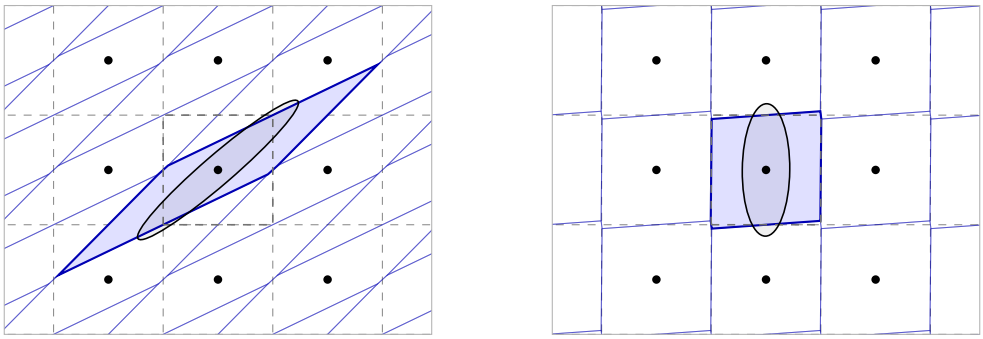
Matrix norms and the condition number.

$$\|\boldsymbol{A}\|_2 = \sigma_{\max}(\boldsymbol{A}), \qquad \|\boldsymbol{A}\|_F = \Big(\sum_{i,j} a_{ij}^2\Big)^{1/2} = \Big(\sum_i \sigma_i^2\Big)^{1/2}, \qquad \kappa_2(\boldsymbol{A}) = \frac{\sigma_{\max}}{\sigma_{\min}}. \tag{11}$$

The condition number bounds how far relative errors in the data can grow in the solution. In floating point a solve loses roughly $\log_{10}\kappa$ of the 16 significant digits that double precision carries, and forming normal equations squares it: $\kappa_2(\boldsymbol{A}^{\mathrm{T}}\boldsymbol{A}) = \kappa_2(\boldsymbol{A})^2$ [4].

Example 3 (dilution of precision is a matrix norm). With unit weights, $\boldsymbol{P}_{\hat{x}} = (\boldsymbol{G}^{\mathrm{T}}\boldsymbol{G})^{-1} = \boldsymbol{V}\boldsymbol{\Sigma}^{-2}\boldsymbol{V}^{\mathrm{T}}$, so

$$\mathrm{GDOP} = \sqrt{\operatorname{tr}(\boldsymbol{G}^{\mathrm{T}}\boldsymbol{G})^{-1}} = \Big(\sum_i \sigma_i^{-2}\Big)^{1/2} = \|\boldsymbol{G}^{+}\|_F. \tag{12}$$



(a) correlation 0.96 (b) after $\boldsymbol{Z}$, correlation 0.03

Figure 3: Pull-in regions for the two covariances of Example 2, drawn to scale in cycles; dots are integer vectors and the correct one is at the centre. Dashed grey: rounding cells, the unit squares of Equation (9). Blue: integer least-squares cells of Equation (10), the central one shaded. Grey ellipse: the 95 % region of the float solution. In (a) the float solution reaches well outside the central square but only to the edge of the central hexagon. In (b) the two kinds of cell nearly coincide, which is why rounding matches integer least squares after decorrelation.

For Figure 1 the singular values of $\boldsymbol{G}$ are 3.331, 1.668, 1.331 and 0.590, and both sides equal 1.9705. The spectral norm $\|\boldsymbol{G}^{+}\|_2 = 1/\sigma_{\min} = 1.694$ is the largest gain from a unit-norm vector of range errors to position and clock. GDOP averages over all directions; the spectral norm is the worst case. The partial measures are HDOP 0.983, VDOP 1.423 and TDOP 0.944.

Example 4 (a determinant does not measure conditioning). Scaling an $n \times n$ matrix by c gives $\det(c\boldsymbol{N}) = c^n \det \boldsymbol{N}$ but leaves $\kappa(c\boldsymbol{N}) = \kappa(\boldsymbol{N})$. For the code-only normal matrix with five parameters (east, north, up, clock, zenith wet delay, the state of the WLS seed of the Estimator Infrastructure document) and weights $1/\sigma^2$:

σ	$\det \boldsymbol{N}$	$\kappa(\boldsymbol{N})$
0.3 m	2.8×10^{6}	1.93×10^{3}
1 m	16.4	1.93×10^{3}
3 m	2.8×10^{-4}	1.93×10^{3}
5 m	1.7×10^{-6}	1.93×10^{3}

A guard that rejects the solution when the determinant falls below a fixed threshold, such as the determinant guard of the WLS seed, responds to the variance units and not to the geometry: the same satellites pass easily at 0.3 m and approach the threshold as the variances grow. A scale-invariant rank test compares each singular value with the largest (for a symmetric positive semidefinite matrix, the eigenvalues) and counts those below $\tau\,\sigma_{\max}$. The table also prices the zenith delay: $\kappa(\boldsymbol{G}^{\mathrm{T}}\boldsymbol{G})$ is 31.8 for position and clock alone and 1933 once the delay joins them, because its mapping function (roughly $1/\sin e$) is one more function of elevation that the up and clock columns nearly reproduce.

1.3 Hermitian matrices

A matrix is Hermitian when it equals its conjugate transpose, $\boldsymbol{A} = \boldsymbol{A}^{\mathsf{H}}$. For real matrices that is symmetry, and every covariance, weight, normal and Kalman covariance matrix in GNSS estimation

is real symmetric. The spectral theorem gives

$$\boldsymbol{Q} = \boldsymbol{V}\boldsymbol{\Lambda}\boldsymbol{V}^{\mathrm{T}}, \qquad \boldsymbol{V}^{\mathrm{T}}\boldsymbol{V} = \boldsymbol{I}, \qquad \lambda_i \in \mathbb{R}, \tag{13}$$

and a covariance matrix is positive semidefinite, $\lambda_i \geq 0$. The eigenvalues bound the variance in any direction $\boldsymbol{d}$ (the Rayleigh quotient):

$$\lambda_{\min} \leq \frac{\boldsymbol{d}^{\mathrm{T}}\boldsymbol{Q}\,\boldsymbol{d}}{\boldsymbol{d}^{\mathrm{T}}\boldsymbol{d}} \leq \lambda_{\max}. \tag{14}$$

Complex Hermitian matrices appear one level down, in the receiver's signal processing (Example 7).

Example 5 (an error ellipse is an eigendecomposition). Code-only solution for Figure 1 with $\sigma = 0.3$ m. The horizontal block of $\boldsymbol{P}_{\hat{x}} = \sigma^2(\boldsymbol{G}^{\mathrm{T}}\boldsymbol{G})^{-1}$ is

$$\boldsymbol{Q}_{EN} = \begin{bmatrix} 0.0337 & 0.0053 \\ 0.0053 & 0.0533 \end{bmatrix} \ \mathrm{m}^2.$$

Its eigenvalues, 0.0547 and 0.0324 m^2, give 1σ semi-axes of 0.234 and 0.180 m. A 95 % region in two dimensions scales them by $\sqrt{\chi^2_2(0.95)} = 2.448$, to 0.572 by 0.440 m, with the major axis along the first eigenvector at azimuth 14.1°. The vertical standard deviation is larger, 0.427 m, for the reason Example 8 gives.

Cholesky: whitening, simulation, decorrelation. A symmetric positive definite matrix factors as $\boldsymbol{Q} = \boldsymbol{L}\boldsymbol{L}^{\mathrm{T}}$ with $\boldsymbol{L}$ lower triangular. The factor is used for

- solving the normal equations, at half the cost of an LU factorisation and without pivoting;
- whitening correlated observations: $\boldsymbol{L}^{-1}\boldsymbol{y}$ has identity covariance (the network estimator whitens its correlated ionosphere-constraint and extra-wide-lane row blocks this way, in `networkWriteWhitenedRows`, `src/core/network/assembly/StationBuild.cpp`);
- simulating correlated noise: $\boldsymbol{\varepsilon} = \boldsymbol{L}\boldsymbol{z}$ with $\boldsymbol{z} \sim N(\boldsymbol{0}, \boldsymbol{I})$;
- in the form $\boldsymbol{Q}_{\hat{N}} = \boldsymbol{L}^{\mathrm{T}}\boldsymbol{D}\boldsymbol{L}$, the LAMBDA decorrelation and search (the LDL^{T} factorisation of the Ambiguity Resolution document).

Example 6 (double differences share a reference satellite). With undifferenced variance σ^2 on every phase observation, n double differences (the double-difference section of the Foundations document) against one reference satellite have

$$\boldsymbol{Q}_{\nabla\Delta} = 2\sigma^2\big(\boldsymbol{I}_n + \boldsymbol{1}\boldsymbol{1}^{\mathrm{T}}\big), \qquad \boldsymbol{Q}^{-1}_{\nabla\Delta} = \frac{1}{2\sigma^2}\Big(\boldsymbol{I}_n - \frac{\boldsymbol{1}\boldsymbol{1}^{\mathrm{T}}}{n+1}\Big), \tag{15}$$

the inverse following from the Sherman–Morrison formula. The eigenvalues are $2\sigma^2$ with multiplicity $n-1$ and $2\sigma^2(n+1)$ along $\boldsymbol{1}$. For eight satellites ($n = 7$) and $\sigma = 3$ mm they are 18 mm^2 six times and 144 mm^2 once. The reference satellite's error enters every double difference, so the common-mode direction carries eight times the variance of any other. Treating the double differences as independent, each with its true variance $4\sigma^2$, overweights that common mode by a factor of 4 and underweights every other direction by a factor of 2. The inverse above, or whitening by the Cholesky factor, weights them correctly.

Example 7 (antenna arrays: a complex Hermitian covariance). An array of M elements sampling complex baseband (I/Q) snapshots $\boldsymbol{s}(t) \in \mathbb{C}^M$ has the spatial covariance

$$\boldsymbol{C}_s = \mathsf{E}\{\boldsymbol{s}\,\boldsymbol{s}^{\mathsf{H}}\} = \sum_k p_k\,\boldsymbol{\alpha}(\theta_k)\,\boldsymbol{\alpha}(\theta_k)^{\mathsf{H}} + \sigma_n^2 \boldsymbol{I}, \tag{16}$$

which is Hermitian and positive definite; $\boldsymbol{\alpha}(\theta)$ is the steering vector of a source at angle θ and p_k its power. GNSS signals arrive 20 to 30 dB below the thermal noise, so the eigenvectors with large eigenvalues belong to interference. Projecting onto their orthogonal complement, $\boldsymbol{\Pi}^{\perp} = \boldsymbol{I} - \boldsymbol{U}_J \boldsymbol{U}_J^{\mathsf{H}}$, removes the interference without knowing where the satellites are [5]. With four elements at half-wavelength spacing, a jammer at 40° with a jammer-to-noise ratio of 40 dB, a satellite at −15° with a signal-to-noise ratio of −20 dB, and 4000 snapshots:

- the sample covariance has eigenvalues 0.98, 1.00, 1.03 and 39 648 (noise power 1). One stands out, near M times the jammer power; the satellite does not show in the spectrum at all;
- the projection attenuates the jammer direction by 77.9 dB and costs the satellite direction 0.09 dB.

Array weights	Output signal-to-jammer-plus-noise ratio
Single element	−60.0 dB
Beam steered at the satellite	−43.3 dB
Projection, then beam	−14.1 dB
MVDR, $\boldsymbol{w} \propto \boldsymbol{C}_s^{-1}\boldsymbol{\alpha}(\theta_{\text{sat}})$	−14.1 dB

The last two recover the full array gain of a clean sky, $-20\ \text{dB} + 10\log_{10} 4 = -14.0$ dB. Correlation with the spreading code then lifts the signal above the noise as usual.

Keeping a Kalman covariance symmetric and positive definite. Rounding error erodes symmetry and positive definiteness in the short-form update $(\boldsymbol{I}-\boldsymbol{K}\boldsymbol{H})\boldsymbol{P}$, most of all when states with millimetre-level uncertainty (converged ambiguities) share the matrix with states whose uncertainty is kilometres (a freshly initialised clock). The engine uses the Joseph form and symmetrises the result (the Joseph-form update of the Kalman filter document). Square-root and UD filters go further: they carry a factor $\boldsymbol{S}$ with $\boldsymbol{P} = \boldsymbol{S}\boldsymbol{S}^{\mathrm{T}}$ and $\kappa(\boldsymbol{S}) = \sqrt{\kappa(\boldsymbol{P})}$ [2]. These, and not the pseudoinverse, are the tools for a filter whose code and phase precisions differ by three orders of magnitude.

1.4 The singular value decomposition

Every real $m \times n$ matrix factors as

$$\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^{\mathrm{T}} = \sum_{i=1}^{r} \sigma_i\,\boldsymbol{u}_i \boldsymbol{v}_i^{\mathrm{T}}, \qquad \sigma_1 \ge \sigma_2 \ge \cdots \ge \sigma_r > 0, \tag{17}$$

with orthonormal $\boldsymbol{u}_i$ in observation space and $\boldsymbol{v}_i$ in parameter space. A parameter change along $\boldsymbol{v}_i$ moves the observations by σ_i along $\boldsymbol{u}_i$. The σ_i^2 are the eigenvalues of $\boldsymbol{A}^{\mathrm{T}}\boldsymbol{A}$, and for a symmetric positive semidefinite matrix the eigenvalues and singular values coincide. The numerical rank is the number of σ_i above a tolerance $\tau\sigma_1$. The $\boldsymbol{v}_i$ with $\sigma_i \approx 0$ span the null space: parameter combinations the data cannot see. In a network that estimates every receiver and satellite clock, adding one constant to all of them leaves every observation unchanged, so a zero singular value is structural there (Example 13; the two-receiver worked example of the Network-mode document). The network estimator counts such directions from the eigenvalues of its reduced normals before every solve (the S-basis section of the Network-mode document).

Example 8 (the weakest direction of the geometry). The right singular vector for the smallest singular value of $\boldsymbol{G}$ (0.590) is

$$\boldsymbol{v}_4 = (-0.03,\ -0.14,\ 0.83,\ 0.54) \quad \text{in (east, north, up, clock).}$$

Raising the antenna by 0.83 m while advancing the clock by 0.54 m changes the eight ranges by 0.32 m at most. The clock column is all ones and the up column is $-\sin e_i$, negative for every satellite above the horizon, so the two columns are close to parallel: the up and clock estimates correlate at +0.93, and VDOP (1.42) exceeds HDOP (0.98). Satellites below the horizon would break the near-parallelism, which is why ground-based GNSS measures height worse than horizontal position.

Example 9 (attitude from antenna baselines). Three antennas on a vehicle give two baselines with known body-frame vectors $\boldsymbol{b}_k$ (1.5 m forward and 1.0 m to the side) and RTK-measured local-frame vectors $\boldsymbol{r}_k$. The attitude is the rotation that minimises $\sum_k \|\boldsymbol{r}_k - \boldsymbol{C}\boldsymbol{b}_k\|^2$ (Wahba's problem [12]). With $\boldsymbol{B} = \sum_k \boldsymbol{r}_k \boldsymbol{b}_k^{\mathrm{T}} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^{\mathrm{T}}$ [6],

$$\hat{\boldsymbol{C}} = \boldsymbol{U}\ \mathrm{diag}\big(1,\ 1,\ \det(\boldsymbol{U}\boldsymbol{V}^{\mathrm{T}})\big)\ \boldsymbol{V}^{\mathrm{T}}. \tag{18}$$

An unconstrained 3×3 fit would have nine unknowns and six equations; the rotation constraint makes the problem well posed, and the SVD solves it in closed form. With 3 mm noise on each baseline component over 5000 trials, the RMS errors are 0.095° in yaw, 0.116° in pitch and 0.171° in roll. Each error follows its baseline: roll depends on the 1.0 m side baseline (3 mm over 1.0 m is 0.17°) and pitch on the 1.5 m forward one (0.11°). $\boldsymbol{B}$ has rank 2 (singular values 2.25, 0.99 and 0), and the determinant term fixes the third axis.

The same step keeps an inertial attitude matrix orthonormal. Replacing a drifted attitude matrix $\boldsymbol{C}$ (the $\boldsymbol{C}_b^e$ of the inertial path) by $\boldsymbol{U}\boldsymbol{V}^{\mathrm{T}}$ from its SVD gives the nearest orthogonal matrix in the Frobenius norm: in the example $\|\boldsymbol{C}^{\mathrm{T}}\boldsymbol{C} - \boldsymbol{I}\|_F$ drops from 5.2×10^{-3} to 9×10^{-16}. The engine applies it during strapdown integration and after every attitude feedback (the strapdown section of the Inertial Aiding document).

Example 10 (common-mode error in a regional network). Daily coordinate residuals of the stations in a regional network share a common-mode signal from reference-frame, large-scale atmospheric and loading errors. With the residual matrix $\boldsymbol{X}$ (days × stations, column means removed), the leading term $\sigma_1 \boldsymbol{u}_1 \boldsymbol{v}_1^{\mathrm{T}}$ of its SVD estimates that signal: $\boldsymbol{u}_1$ carries its time series and $\boldsymbol{v}_1$ each station's response [3]. The example is synthetic: up component, 12 stations, 730 days, a time-correlated common signal of 2.5 mm scaled by a station response between 0.8 and 1.2, and 2.0 mm of white noise per station.

- The first component carries 66.1 % of the variance; the next two, 3.9 % and 3.6 %.
- Removing it lowers the mean station RMS from 3.27 mm to 1.91 mm, the white-noise level.
- The recovered series correlates 0.977 with the true common mode, and $\boldsymbol{v}_1$ correlates 0.962 with the true station responses.

Ill-posed inversions: truncated SVD and Tikhonov. Ionospheric and water-vapour tomography lead to systems whose singular values decay toward zero with no clear gap. The least-squares solution

$$\hat{\boldsymbol{x}} = \sum_i \frac{\boldsymbol{u}_i^{\mathrm{T}}\boldsymbol{y}}{\sigma_i}\,\boldsymbol{v}_i \tag{19}$$

then amplifies noise through the small σ_i. Truncated SVD keeps the first k terms; Tikhonov regularisation replaces $1/\sigma_i$ by $\sigma_i/(\sigma_i^2 + \alpha^2)$. Both accept bias to reduce variance, with k or α chosen from the data (the L-curve, generalised cross-validation). That is a deliberate choice for an ill-posed problem, and a different thing from what a default pseudoinverse tolerance does to a well-posed one (Example 11). The engine's SP3 orbit fit solves a well-posed least-squares problem through Equation (19) with a relative cutoff (the sidereal-trigonometric fit of the Orbit and Clock Product Evaluation document).

1.5 The pseudoinverse

The Moore–Penrose pseudoinverse is $\boldsymbol{A}^+ = \boldsymbol{V}\boldsymbol{\Sigma}^+\boldsymbol{U}^{\mathrm{T}}$, where $\boldsymbol{\Sigma}^+$ inverts the nonzero singular values and keeps the zeros. It is the unique matrix that satisfies the four Penrose conditions

$$\boldsymbol{A}\boldsymbol{A}^+\boldsymbol{A} = \boldsymbol{A}, \qquad \boldsymbol{A}^+\boldsymbol{A}\boldsymbol{A}^+ = \boldsymbol{A}^+, \qquad (\boldsymbol{A}\boldsymbol{A}^+)^{\mathrm{T}} = \boldsymbol{A}\boldsymbol{A}^+, \qquad (\boldsymbol{A}^+\boldsymbol{A})^{\mathrm{T}} = \boldsymbol{A}^+\boldsymbol{A}. \tag{20}$$

Three cases matter in GNSS.

- *Full column rank*, as in ordinary positioning: $\boldsymbol{A}^+ = (\boldsymbol{A}^{\mathrm{T}}\boldsymbol{A})^{-1}\boldsymbol{A}^{\mathrm{T}}$, so the pseudoinverse is least squares computed through the SVD. With weights, $\hat{\boldsymbol{x}} = (\boldsymbol{W}^{1/2}\boldsymbol{A})^+\boldsymbol{W}^{1/2}\boldsymbol{y}$; applying $\boldsymbol{A}^+$ to $\boldsymbol{y}$ directly drops the weights.
- *Rank deficient*, as with a datum defect: $\boldsymbol{A}^+\boldsymbol{y}$ is the least-squares solution of minimum norm. It is orthogonal to the null space, which means it chooses a datum without saying so (Example 13; [9]).
- *Numerically rank deficient*: singular values below $\tau\sigma_1$ count as zero. Deliberate truncation regularises an ill-posed problem (Section 1.4); a default tolerance applied to a badly scaled but well-posed problem discards real information (Example 11).

In the engine the pseudoinverse appears in two full-rank roles: the SP3 orbit fit above, and the fallback of the Kalman gain solve when the $\mathrm{LDL}^{\mathrm{T}}$ factorisation of $\boldsymbol{S}$ fails (the measurement-update section of the Kalman filter document).

Example 11 (metre-level code, millimetre-level phase). One epoch of code and phase, Equation (3), with float ambiguities estimated as corrections to a priori values and a position offset of (3, −2, 5) m. The phase-to-code weight ratio is $(\sigma_P/\sigma_\Phi)^2$: 10^4 for the geodetic receiver and 2.25×10^6 for the low-cost one. Each solver column gives the position error against a stable reference, SVD least squares on the whitened system in double precision.

			double precision		single precision	
Case	$\kappa(\boldsymbol{W}^{1/2}\boldsymbol{A})$	$\kappa(\boldsymbol{N})$	Cholesky on $\boldsymbol{N}$	$\boldsymbol{N}^+$	normal eqs.	QR on $\boldsymbol{W}^{1/2}\boldsymbol{A}$
Geodetic	1.8×10^3	3.4×10^6	< 1 µm	< 1 µm	8.3 cm	7 µm
Low-cost	2.8×10^4	7.6×10^8	< 1 µm	< 1 µm	**42 m**	3 µm
Clock in seconds	4.6×10^{11}	2.1×10^{23}	< 1 µm	**5.1 m**	6.9 cm	2 µm

- *Not near-singular.* Double precision carries about 16 digits, and $\kappa(\boldsymbol{N})$ between 10^6 and 10^9 leaves 7 to 10 of them. The weight matrix is diagonal, so its inverse is exact however wide its range.
- *The weight range costs digits through the normal equations*, because $\kappa(\boldsymbol{N}) = \kappa(\boldsymbol{W}^{1/2}\boldsymbol{A})^2$. In single precision that is the difference between 42 m and 3 µm in the low-cost case. QR on the whitened system never forms $\boldsymbol{N}$.

- *Units, not information.* A clock parameter in seconds multiplies its column by $c = 2.998 \times 10^8$ m/s and drives $\kappa(\boldsymbol{N})$ to 2.1×10^{23}. Cholesky is insensitive to this kind of diagonal scaling and still returns the position within 1 µm. The pseudoinverse of $\boldsymbol{N}$ with numpy's default relative tolerance ($10^{-15}\sigma_1$) sets 11 of its 12 singular values to zero and returns the a priori position unchanged: a 5.1 m error with no warning.

The remedies are consistent units (the clock in metres, unknowns as corrections to a priori values), whitening followed by QR or by Cholesky in double precision, and square-root forms in filters. The pseudoinverse is not a remedy for ill-conditioning.

Example 12 (what phase adds to a single epoch). Order the parameters of Equation (3) as the position and clock $\boldsymbol{x}$ followed by the ambiguities $N_1, \ldots, N_m$, with $\boldsymbol{W}_P = \sigma_P^{-2}\boldsymbol{I}$ and $\boldsymbol{W}_\Phi = \sigma_\Phi^{-2}\boldsymbol{I}$. The normal matrix is

$$\boldsymbol{N} = \begin{bmatrix} \boldsymbol{N}_{xx} & \boldsymbol{N}_{xa} \\ \boldsymbol{N}_{ax} & \boldsymbol{N}_{aa} \end{bmatrix} = \begin{bmatrix} \boldsymbol{G}^{\mathrm{T}}(\boldsymbol{W}_P + \boldsymbol{W}_\Phi)\,\boldsymbol{G} & \lambda_{\mathrm{L1}}\,\boldsymbol{G}^{\mathrm{T}}\boldsymbol{W}_\Phi \\ \lambda_{\mathrm{L1}}\,\boldsymbol{W}_\Phi\boldsymbol{G} & \lambda_{\mathrm{L1}}^2\,\boldsymbol{W}_\Phi \end{bmatrix}. \tag{21}$$

Eliminating the ambiguities (the Schur complement) leaves

$$\boldsymbol{N}_{xx} - \boldsymbol{N}_{xa}\boldsymbol{N}_{aa}^{-1}\boldsymbol{N}_{ax} = \boldsymbol{G}^{\mathrm{T}}\boldsymbol{W}_P\,\boldsymbol{G}, \tag{22}$$

the code-only normal matrix. With float ambiguities, one epoch of phase adds nothing to the position: in Example 11 the code-plus-phase and code-only positions agree to 3×10^{-13} m. This is the single-receiver case of the statement in the redundancy section of the Stochastic Model and QC document that per-arc float ambiguities leave no phase-side redundancy. Phase precision reaches the position only once the ambiguities are fixed to integers or tied together over time.

The weight range reappears in the float ambiguities:

$$\boldsymbol{Q}_{\hat{N}} = \frac{1}{\lambda_{\mathrm{L1}}^2}\Big(\sigma_\Phi^2\,\boldsymbol{I}_m + \sigma_P^2\,\boldsymbol{\Pi}_G\Big), \qquad \boldsymbol{\Pi}_G = \boldsymbol{G}(\boldsymbol{G}^{\mathrm{T}}\boldsymbol{G})^{-1}\boldsymbol{G}^{\mathrm{T}}. \tag{23}$$

$\boldsymbol{\Pi}_G$ is an orthogonal projector of rank 4, so the eigenvalues of $\boldsymbol{Q}_{\hat{N}}$ are $(\sigma_P^2 + \sigma_\Phi^2)/\lambda_{\mathrm{L1}}^2$ four times and $\sigma_\Phi^2/\lambda_{\mathrm{L1}}^2$ the remaining $m - 4$ times, and

$$\boxed{\kappa(\boldsymbol{Q}_{\hat{N}}) = 1 + \frac{\sigma_P^2}{\sigma_\Phi^2}.} \tag{24}$$

Computed: eigenvalues of 2.486 and 2.49×10^{-4} cycles^2 and $\kappa = 1.0001 \times 10^4$ in the geodetic case, and $\kappa = 2.25 \times 10^6$ in the low-cost case, both matching Equation (24). The ambiguity search space is an ellipsoid 100 times longer than it is wide in the geodetic case and 1500 times in the low-cost case. That elongation is why LAMBDA decorrelates before it searches (Example 2).

Example 13 (a network with a datum defect). Three receivers track eight satellites, and the only parameters are clocks: $y_{ri} = \mathrm{d}t_r - \mathrm{d}t^i + \varepsilon_{ri}$ in metres, with $\sigma = 1$ cm. Adding the same constant to every clock leaves every observation unchanged, so $\boldsymbol{A}\mathbf{1} = \mathbf{0}$: rank 10 for 11 parameters, with a smallest singular value of 3×10^{-16}. This is the null vector of the two-receiver worked example of the Network-mode document, which compares two pinned datums. Here the pseudoinverse is the second solution:

- *Pseudoinverse*: minimum norm, hence $\mathbf{1}^{\mathrm{T}}\hat{\boldsymbol{x}}^{+} = 0$; the datum is that the eleven clocks sum to zero.

- *Explicit S-basis*: the first receiver's clock is fixed to zero (the pivot) and the remaining full-rank system is solved.

Quantity	Pseudoinverse	Pivot
Satellite clock differences $\mathrm{d}t^i - \mathrm{d}t^1$ (estimable)	agree to 2×10^{-10} m	
Receiver clocks (not estimable)	23.6, −128.1, 51.7 km	0, −151.7, 28.1 km
Trace of the covariance (unit variance)	2.674	3.833
Change in the receiver clocks when the +468 µs satellite sets	+13 534.8 m (45.15 µs), all three	0, 0.8, −7.7 mm

The pseudoinverse has the smaller trace, the minimum-trace property of free-network adjustment, but it does not remove the defect. It picks a datum, and that datum moves whenever the parameter set changes. For a departing parameter x_k the jump is $(x_k - \bar{x})/(n-1)$, with $\bar{x}$ the mean of the n parameters before it left: here $(468.0 - 16.5)/10 = 45.15$ µs, as computed. GNSS networks and PPP-RTK therefore choose the S-basis explicitly: a pivot receiver, a pivot satellite, or a zero-mean condition with rules for set changes [7]. Each estimable combination then has a stated meaning (the S-basis section of the Network-mode document).

Choosing a solver.

Situation	Tool
Full rank, well scaled	Cholesky on $\boldsymbol{N}$ in double precision, or QR on $\boldsymbol{W}^{1/2}\boldsymbol{A}$
Wide weight range, single precision, long filters	Whitening and QR; a square-root or UD filter for sequential updates
Mixed units (seconds, absolute coordinates)	Rescaling: the clock in metres, unknowns as corrections to a priori values
Datum defect	An explicit S-basis; the pseudoinverse only when estimable functions are all that matter
Ill-posed inversion (tomography)	Truncated SVD or Tikhonov with a data-driven parameter
Rank diagnosis	Singular values or eigenvalues relative to the largest, never the determinant

1.6 Exercises

1. Show that GDOP $= \|\boldsymbol{G}^{+}\|_F$ and $\|\boldsymbol{G}^{+}\|_2 = 1/\sigma_{\min}$ for a geometry matrix of full column rank (Example 3). Which of the two answers a worst-case question?

2. Form the partitioned normal matrix Equation (21), eliminate the ambiguities, and show that the reduced normal matrix equals $\boldsymbol{G}^{\mathrm{T}}\boldsymbol{W}_P\boldsymbol{G}$ (Example 12). Explain what changes once the ambiguities are fixed to integers.

3. Derive Equation (23) and the eigenvalues of $\boldsymbol{Q}_{\hat{N}}$, and show Equation (24).

4. For $\boldsymbol{Q}_{\nabla\Delta}$ of Equation (15), find the eigenvalues and eigenvectors and verify the Sherman–Morrison inverse (Example 6). Whiten simulated double differences with the Cholesky factor and check that their sample covariance is close to the identity.

5. Show that the pseudoinverse solution of Example 13 satisfies $\mathbf{1}^{\mathrm{T}}\hat{\boldsymbol{x}}^{+} = 0$, and derive the jump $(x_k - \bar{x})/(n-1)$ when parameter k leaves.

6. In the third case of Example 11 the clock column is scaled by c. Show that column scaling changes κ but not the least-squares position, then explain why the pseudoinverse of $\boldsymbol{N}$ failed while Cholesky did not. (The pseudoinverse tolerance is relative to σ_1.)

7. The WLS seed rejects a solution through the determinant guard of the Estimator Infrastructure document (`WlsEstimator::estimate`). Use Example 4 to explain what this test measures, and replace it with a scale-invariant test.

8. Show that maximising $\operatorname{tr}(\boldsymbol{C}^{\mathrm{T}}\boldsymbol{B})$ over rotations gives Equation (18), and explain why two baselines suffice although $\boldsymbol{B}$ has rank 2 (Example 9).

9. Move the multipath error of Example 1 from S4 to S7. Predict the change from the redundancy numbers, then run the script to check.

1.7 Reproducing the numbers and further reading

Every number in Examples 1 to 13 is printed by

```
python scripts/linalg_supplement_examples.py
```

which needs only numpy and runs in under a minute. Each example has its own random seed, so the output is reproducible and one example can be changed without disturbing the others.

The same operations appear in the engine as follows.

Operation	Symbol (file)
Cholesky whitening of correlated row blocks into unit-information rows	`networkWriteWhitenedRows` (`src/core/network/assembly/StationBuild.cpp`), called by the ionosphere-constraint rows (`src/core/network/assembly/IonoConstraintRows.cpp`); the single-difference extra-wide-lane rows whiten their own groups (`src/core/network/ar/SdEwlRows.cpp`)
Condition number of the WLS normal matrix (logged through an SVD) and the determinant guard	`WlsEstimator::estimate` (`src/core/math/WlsEstimator.cpp`)
LDL^{T} solve of $\boldsymbol{S}$ with a pseudoinverse fallback; Joseph update; SVD re-orthonormalisation after attitude feedback	`KalmanFilter::update` (`src/core/filter/KalmanFilter.cpp`)
SVD re-orthonormalisation during strapdown integration	`ImuIntegrator::integrate` (`src/core/filter/ImuIntegrator.cpp`)
SP3 orbit fit by truncated-SVD least squares	`Interpolate::trigSvdInterp` (`src/core/math/Interpolate.cpp`)
Null-space (datum) count from eigenvalues	`NetworkSbasis::assertNullSpaceRank` (`src/core/network/solve/Sbasis.cpp`)
LDL^{T} decorrelation and integer search (MLAMBDA)	`src/core/filter/IlsCore.cpp`, `AmbiguityResolver` (`src/core/filter/AmbiguityResolver.cpp`)

For further reading, Strang [8] covers norms, the SVD and the pseudoinverse at the level of the course; Higham [4] covers conditioning, normal equations against QR, and scaling; Teunissen [11] treats weighted least squares, redundancy and S-bases in geodetic notation; and Odijk et al. [7] apply S-system theory to undifferenced PPP-RTK networks.

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